Mapping MeDIP counts matrix to Glioma SOLID patients

April 14th, 2026

Set up

```
library(data.table) |> suppressMessages()
```

```
## Warning: package 'data.table' was built under R version 4.5.2
```

```
dir <- "/Users/julianguyen/Documents/SOLID-methylation/"
```

Load in data

MeDIP log2CPM matrix

```
medip <- readRDS(paste0(dir, "data/rawdata/MeDIP_Solid_all_log2CPM.rds"))
print(dim(medip))
```

```
## [1] 505027    174
```

```
medip[1:5,1:5]
```

```
##                       batch7B_13 batch7B_37 batch7B_38 batch7B_39  batch7B_40
## chr1.941701.942000   -19.931569 -2.0409658 -1.0770380 -0.6734604 -1.36959080
## chr1.942001.942300    -1.988499 -0.9629665  0.2485361  0.3265385  0.10701959
## chr1.945601.945900    -2.988493 -1.5864016 -1.6126330 -1.2209471 -1.22094708
## chr1.945901.946200   -19.931569 -3.0588817 -4.0588696 -1.9213847 -1.53532755
## chr1.1008601.1008900  -2.988493 -1.0499019 -0.5479297 -0.1140337 -0.07703951
```

```
available_medip <- colnames(medip)
available_medip
```

```
##   [1] "batch7B_13"         "batch7B_37"         "batch7B_38"
##   [4] "batch7B_39"         "batch7B_40"         "batch7B_41"
##   [7] "batch7B_14"         "batch7B_42"         "batch7B_43"
##  [10] "batch7B_44"         "batch7B_45"         "batch7B_46"
##  [13] "batch7B_34"         "batch7B_35"         "batch7B_97"
##  [16] "batch7B_62"         "batch7B_99"         "batch7B_47"
##  [19] "batch7B_36"         "batch7B_48"         "batch7B_15"
##  [22] "batch7B_49"         "batch7B_50"         "batch7B_51"
##  [25] "batch7B_52"         "batch6_Solid_4-1"   "batch6_Solid_4-2"
```

```
##  [28] "batch6_Solid_5-1"   "batch6_Solid_5-2"   "batch6_Solid_5-3"
##  [31] "batch6_Solid_7-1"   "batch6_Solid_7-2"   "batch6_Solid_8-1"
##  [34] "batch6_Solid_8-2"   "batch6_Solid_8-3"   "batch6_Solid_8-4"
##  [37] "batch6_Solid_8-5"   "batch6_Solid_9-1"   "batch6_Solid_9-2B"
##  [40] "batch6_Solid_9-3"   "batch6_Solid_9-4"   "batch6_Solid_9-5"
##  [43] "batch6_Solid_9-6"   "batch6_Solid_9-7"   "batch7B_16"
##  [46] "batch7B_53"         "batch7B_54"         "batch7B_55"
##  [49] "batch7B_100"        "batch6_Solid_11-1"  "batch6_Solid_11-2"
##  [52] "batch6_Solid_11-3"  "batch6_Solid_11-4"  "batch7B_101"
##  [55] "batch7B_68"         "batch6_Solid_13-1"  "batch6_Solid_13-2"
##  [58] "batch6_Solid_13-3"  "batch6_Solid_13-4"  "batch6_Solid_13-5"
##  [61] "batch6_Solid_13-6"  "batch6_Solid_13-7"  "batch6_Solid_13-8"
##  [64] "batch6_Solid_13-9"  "batch6_Solid_13-10" "batch6_Solid_13-11"
##  [67] "batch6_Solid_14-1"  "batch6_Solid_14-2C" "batch6_Solid_14-3"
##  [70] "batch6_Solid_14-4C" "batch6_Solid_15-1"  "batch6_Solid_15-2"
##  [73] "batch6_Solid_15-3"  "batch7B_102"        "batch7B_71"
##  [76] "batch7B_72"         "batch7B_96"         "batch7B_103"
##  [79] "batch7B_75"         "batch7B_106"        "batch7B_107"
##  [82] "batch7B_78"         "batch7B_108"        "batch7B_115"
##  [85] "batch7B_116"        "batch7B_83"         "batch7B_91"
##  [88] "batch6_Solid_19-1"  "batch6_Solid_19-2"  "batch6_Solid_19-3"
##  [91] "batch6_Solid_19-4"  "batch7B_119"        "batch7B_125"
##  [94] "batch7B_124"        "batch7B_122"        "batch7B_121"
##  [97] "batch7B_120"        "batch7B_135"        "batch7B_134"
## [100] "batch7B_133"        "batch7B_126"        "batch7B_140"
## [103] "batch7B_139"        "batch7B_137"        "batch7B_136"
## [106] "batch7B_144"        "batch7B_143"        "batch7B_142"
## [109] "batch7B_141"        "batch7B_146"        "batch7B_145"
## [112] "batch7B_85"         "batch7B_86"         "batch7B_87"
## [115] "batch7B_88"         "batch7B_89"         "batch7B_90"
## [118] "batch7B_92"         "batch7B_104"        "batch7B_95"
## [121] "batch7B_105"        "batch7B_17"         "batch7B_182"
## [124] "batch7B_183"        "batch7B_80"         "batch7B_18"
## [127] "batch7B_147"        "batch7B_148"        "batch7B_152"
## [130] "batch7B_157"        "batch7B_149"        "batch7B_207"
## [133] "batch7B_151"        "batch7B_153"        "batch7B_184"
## [136] "batch7B_171"        "batch7B_154"        "batch7B_169"
## [139] "batch7B_19"         "batch7B_158"        "batch7B_156"
## [142] "batch7B_155"        "batch7B_20"         "batch7B_161"
## [145] "batch7B_160"        "batch7B_159"        "batch7B_166"
## [148] "batch7B_165"        "batch7B_164"        "batch7B_170"
## [151] "batch7B_162"        "batch7B_168"        "batch7B_167"
## [154] "batch7B_163"        "batch7B_21"         "batch7B_185"
## [157] "batch7B_186"        "batch7B_188"        "batch7B_189"
## [160] "batch7B_190"        "batch7B_191"        "batch7B_22"
## [163] "batch7B_23"         "batch7B_24"         "batch7B_174"
## [166] "batch7B_173"        "batch7B_175"        "batch7B_59"
## [169] "batch7B_60"         "batch7B_176"        "batch7B_177"
## [172] "batch7B_172"        "batch7B_179"        "batch7B_178"
```

These are the column names of the MeDIP counts matrix for all SOLID patients. There were 174 samples.

Cholangio MeDIP log2CPM matrix

```
cmat <- readRDS(paste0(dir, "data/rawdata/cholangio/MeDIP_Solid_cholangiocarcinoma_log2CPM.rds"))
print(dim(cmat))
```

```
## [1] 505027     56
```

```
cholangio_medip <- colnames(cmat)
cholangio_medip
```

```
##  [1] "batch7B_13"  "batch7B_37"  "batch7B_38"  "batch7B_39"  "batch7B_40"
##  [6] "batch7B_41"  "batch7B_14"  "batch7B_42"  "batch7B_43"  "batch7B_44"
## [11] "batch7B_45"  "batch7B_46"  "batch7B_34"  "batch7B_35"  "batch7B_97"
## [16] "batch7B_62"  "batch7B_99"  "batch7B_47"  "batch7B_36"  "batch7B_48"
## [21] "batch7B_15"  "batch7B_49"  "batch7B_50"  "batch7B_51"  "batch7B_52"
## [26] "batch7B_16"  "batch7B_53"  "batch7B_54"  "batch7B_55"  "batch7B_100"
## [31] "batch7B_101" "batch7B_68"  "batch7B_102" "batch7B_71"  "batch7B_72"
## [36] "batch7B_96"  "batch7B_103" "batch7B_75"  "batch7B_106" "batch7B_107"
## [41] "batch7B_78"  "batch7B_108" "batch7B_115" "batch7B_116" "batch7B_83"
## [46] "batch7B_91"  "batch7B_85"  "batch7B_86"  "batch7B_87"  "batch7B_88"
## [51] "batch7B_89"  "batch7B_90"  "batch7B_92"  "batch7B_104" "batch7B_95"
## [56] "batch7B_105"
```

These should be all the cholangio patients. There are 56 samples total. This leaves 118 (174-56) MeDIP samples from the earlier matrix to be from the glioma patients.

Identify Glioma MeDIP IDs

Removing the cholangio MeDIP sample IDs, these are what we are left with for glioma patients.

```
glioma_medip <- available_medip[-which(available_medip %in% cholangio_medip)]
glioma_medip
```

```
##   [1] "batch6_Solid_4-1"   "batch6_Solid_4-2"   "batch6_Solid_5-1"
##   [4] "batch6_Solid_5-2"   "batch6_Solid_5-3"   "batch6_Solid_7-1"
##   [7] "batch6_Solid_7-2"   "batch6_Solid_8-1"   "batch6_Solid_8-2"
##  [10] "batch6_Solid_8-3"   "batch6_Solid_8-4"   "batch6_Solid_8-5"
##  [13] "batch6_Solid_9-1"   "batch6_Solid_9-2B"  "batch6_Solid_9-3"
##  [16] "batch6_Solid_9-4"   "batch6_Solid_9-5"   "batch6_Solid_9-6"
##  [19] "batch6_Solid_9-7"   "batch6_Solid_11-1"  "batch6_Solid_11-2"
##  [22] "batch6_Solid_11-3"  "batch6_Solid_11-4"  "batch6_Solid_13-1"
##  [25] "batch6_Solid_13-2"  "batch6_Solid_13-3"  "batch6_Solid_13-4"
##  [28] "batch6_Solid_13-5"  "batch6_Solid_13-6"  "batch6_Solid_13-7"
##  [31] "batch6_Solid_13-8"  "batch6_Solid_13-9"  "batch6_Solid_13-10"
##  [34] "batch6_Solid_13-11" "batch6_Solid_14-1"  "batch6_Solid_14-2C"
##  [37] "batch6_Solid_14-3"  "batch6_Solid_14-4C" "batch6_Solid_15-1"
##  [40] "batch6_Solid_15-2"  "batch6_Solid_15-3"  "batch6_Solid_19-1"
##  [43] "batch6_Solid_19-2"  "batch6_Solid_19-3"  "batch6_Solid_19-4"
##  [46] "batch7B_119"        "batch7B_125"        "batch7B_124"
##  [49] "batch7B_122"        "batch7B_121"        "batch7B_120"
```

```
##  [52] "batch7B_135"        "batch7B_134"        "batch7B_133"
##  [55] "batch7B_126"        "batch7B_140"        "batch7B_139"
##  [58] "batch7B_137"        "batch7B_136"        "batch7B_144"
##  [61] "batch7B_143"        "batch7B_142"        "batch7B_141"
##  [64] "batch7B_146"        "batch7B_145"        "batch7B_17"
##  [67] "batch7B_182"        "batch7B_183"        "batch7B_80"
##  [70] "batch7B_18"         "batch7B_147"        "batch7B_148"
##  [73] "batch7B_152"        "batch7B_157"        "batch7B_149"
##  [76] "batch7B_207"        "batch7B_151"        "batch7B_153"
##  [79] "batch7B_184"        "batch7B_171"        "batch7B_154"
##  [82] "batch7B_169"        "batch7B_19"         "batch7B_158"
##  [85] "batch7B_156"        "batch7B_155"        "batch7B_20"
##  [88] "batch7B_161"        "batch7B_160"        "batch7B_159"
##  [91] "batch7B_166"        "batch7B_165"        "batch7B_164"
##  [94] "batch7B_170"        "batch7B_162"        "batch7B_168"
##  [97] "batch7B_167"        "batch7B_163"        "batch7B_21"
## [100] "batch7B_185"        "batch7B_186"        "batch7B_188"
## [103] "batch7B_189"        "batch7B_190"        "batch7B_191"
## [106] "batch7B_22"         "batch7B_23"         "batch7B_24"
## [109] "batch7B_174"        "batch7B_173"        "batch7B_175"
## [112] "batch7B_59"         "batch7B_60"         "batch7B_176"
## [115] "batch7B_177"        "batch7B_172"        "batch7B_179"
## [118] "batch7B_178"
```

Subet for matched glioma SOLID IDs

Here are some metadata files I have available.

Cholangio pheno

```r
pheno <- fread(paste0(dir, "data/rawdata/cholangio/Pheno_cholangio_all_IDHwt_Vs_IDHmut.txt"))
dim(pheno) # 64 samples here
```

```
## [1] 64 15
```

```r
head(pheno)
```

```
##    tube_number solid_patient  phase MV_labels YE_labels tumour_type
##          <int>        <char> <char>    <lgcl>    <char>      <char>
## 1:          NA       HPB1695   <NA>        NA      <NA>       IDHwt
## 2:          NA       HPB1699   <NA>        NA      <NA>       IDHwt
## 3:          NA       HPB1702   <NA>        NA      <NA>       IDHwt
## 4:          NA       HPB1737   <NA>        NA      <NA>       IDHwt
## 5:          NA       HPB1769   <NA>        NA      <NA>       IDHwt
## 6:          NA       HPB1822   <NA>        NA      <NA>       IDHwt
##    medip_processed blood_collection_date closest_MRI response taking_drug
##             <char>                <char>      <char>   <char>      <char>
## 1:           yosef                  <NA>        <NA> baseline          no
## 2:           yosef                  <NA>        <NA> baseline          no
## 3:           yosef                  <NA>        <NA> baseline          no
## 4:           yosef                  <NA>        <NA> baseline          no
## 5:           yosef                  <NA>        <NA> baseline          no
```

```
## 6:           yosef                  <NA>        <NA> baseline          no
##      misc sequenced  MeDIP         MeDIP_ID
##    <char>    <char> <char>           <char>
## 1:   <NA>       yes    yes batch9YE_HPB1695
## 2:   <NA>       yes    yes batch9YE_HPB1699
## 3:   <NA>       yes    yes batch9YE_HPB1702
## 4:   <NA>       yes    yes batch9YE_HPB1737
## 5:   <NA>       yes    yes batch9YE_HPB1769
## 6:   <NA>       yes    yes batch9YE_HPB1822
```

The MeDIP_ID column matches the column names, but only the cholangio matrices. I don't have an equivalent metadata file for the glioma samples.

```r
table(pheno$MeDIP_ID %in% available_medip)
```

```
##
## FALSE  TRUE
##     8    56
```

```r
table(pheno$MeDIP_ID %in% glioma_medip)
```

```
##
## FALSE
##    64
```

Glioma pheno

Below is the metadata file from the tissue methylation which has the SOLID ids of the 29 glioma solid patients with methylation. I created a new column `mapping` which can be used to map to the MeDIP IDs.

```r
load(paste0(dir, "data/procdata/RGSets_targets.RData")) |> suppressMessages()
targets$mapping <- gsub("_0", "_", gsub("_0", "_", paste0("Solid_", sub(".*-", "", targets$Subject))))
head(targets)
```

```
##    Sample_Name                                          Basename   Sentrix_ID
## 1   GSM9325970 data/rawdata/idats/GSM9325970_205624630086_R03C01 205624630086
## 3   GSM9325971 data/rawdata/idats/GSM9325971_205624630086_R06C01 205624630086
## 5   GSM9325972 data/rawdata/idats/GSM9325972_205624630086_R08C01 205624630086
## 7   GSM9325975 data/rawdata/idats/GSM9325975_208567280160_R01C01 205624630086
## 9   GSM9325976 data/rawdata/idats/GSM9325976_208567280160_R02C01 205624630086
## 11  GSM9325977 data/rawdata/idats/GSM9325977_208567280160_R03C01 205624630086
##    Sentrix_Position  Array   Subject                 Path_dx
## 1            R03C01   EPIC SOLID-007             Astrocytoma
## 3            R03C01   EPIC SOLID-013             Astrocytoma
## 5            R06C01   EPIC SOLID-015             Astrocytoma
## 7            R06C01 EPICv2 SOLID-024             Astrocytoma
## 9            R08C01 EPICv2 SOLID-032             Astrocytoma
## 11           R08C01 EPICv2 SOLID-040 Oligodendroglial Tumors
##                                                Dx_sub Grade Age    Sex  Race
## 1                                          ASTROCYTOMA     3  41   Male White
## 3      ASTROCYTIC TUMOUR GLIOBLASTOMA (WHO GRADE IV)     4  27   Male White
## 5  IDH MUTANT ANAPLASTIC ASTROCYTOMA (WHO GRADE III)     3  50   Male White
```

```
## 7              ANAPLASTIC ASTROCYTOMA (WHO GRADE IV)     4  28   Male White
## 9                WHO GRADE II FIBRILLARY ASTROCYTOMA     2  39   Male White
## 11                      ANAPLASTIC OLIGODENDROGLIOMA     3  32 Female White
##    ECOG Enroll_Date Off_Tx_Date
## 1     1   01-Apr-20   08-May-20
## 3     1   26-Jun-20   07-Apr-21
## 5     1   21-Jul-20   16-Sep-20
## 7     0   14-Jan-22   09-Nov-22
## 9     1   01-Dec-22   26-Jan-23
## 11    1   13-Feb-23   30-Apr-23
##                                                    OTX_CD OFF_STUDY_DT
## 1  Disease progression, relapse during active treatment    28-May-20
## 3  Disease progression, relapse during active treatment    13-Jul-21
## 5  Disease progression, relapse during active treatment    22-Sep-20
## 7  Disease progression, relapse during active treatment    09-Nov-22
## 9  Disease progression, relapse during active treatment    26-Jan-23
## 11                                                        30-Apr-23
##                         OTX_STUDY_FU_CD Response_RANO PFS_Days      pfs
## 1  Protocol-defined follow-up completed            PD       37 1.216438
## 3  Protocol-defined follow-up completed            PR      285 9.369860
## 5  Protocol-defined follow-up completed            PD       57 1.873972
## 7  Protocol-defined follow-up completed            SD      299 9.830133
## 9                                                  PD       56 1.841095
## 11                                Death            PD       76 2.498629
##    PFS_status   Date_fu OS_status OS_days        os FU_status       idat
## 1           1 07-Oct-22         1     919 30.213687         0 GSM9325970
## 3           1 05-Apr-21         1     283  9.304106         0 GSM9325971
## 5           1 23-Sep-21         1     429 14.104104         0 GSM9325972
## 7           1 16-Jan-24         1     732 24.065744         0 GSM9325975
## 9           1 23-Apr-23         1     143  4.701368         0 GSM9325976
## 11          1 30-Apr-23         1      76  2.498629         0 GSM9325977
##     mapping
## 1   Solid_7
## 3  Solid_13
## 5  Solid_15
## 7  Solid_24
## 9  Solid_32
## 11 Solid_40
```

Method 1: subset for sampleIDs with 'SOLID' explicitly in sampleID

```r
glioma_with_solid <- glioma_medip[grep("Solid", glioma_medip)]
glioma_with_solid
```

```
##  [1] "batch6_Solid_4-1"   "batch6_Solid_4-2"   "batch6_Solid_5-1"
##  [4] "batch6_Solid_5-2"   "batch6_Solid_5-3"   "batch6_Solid_7-1"
##  [7] "batch6_Solid_7-2"   "batch6_Solid_8-1"   "batch6_Solid_8-2"
## [10] "batch6_Solid_8-3"   "batch6_Solid_8-4"   "batch6_Solid_8-5"
## [13] "batch6_Solid_9-1"   "batch6_Solid_9-2B"  "batch6_Solid_9-3"
## [16] "batch6_Solid_9-4"   "batch6_Solid_9-5"   "batch6_Solid_9-6"
## [19] "batch6_Solid_9-7"   "batch6_Solid_11-1"  "batch6_Solid_11-2"
## [22] "batch6_Solid_11-3"  "batch6_Solid_11-4"  "batch6_Solid_13-1"
```

```
## [25] "batch6_Solid_13-2"  "batch6_Solid_13-3"  "batch6_Solid_13-4"
## [28] "batch6_Solid_13-5"  "batch6_Solid_13-6"  "batch6_Solid_13-7"
## [31] "batch6_Solid_13-8"  "batch6_Solid_13-9"  "batch6_Solid_13-10"
## [34] "batch6_Solid_13-11" "batch6_Solid_14-1"  "batch6_Solid_14-2C"
## [37] "batch6_Solid_14-3"  "batch6_Solid_14-4C" "batch6_Solid_15-1"
## [40] "batch6_Solid_15-2"  "batch6_Solid_15-3"  "batch6_Solid_19-1"
## [43] "batch6_Solid_19-2"  "batch6_Solid_19-3"  "batch6_Solid_19-4"
```

```r
# keep only samples with '-1' (to filter for unique and presumably baseline samples)
glioma_with_solid[grepl("-1$", glioma_with_solid)]
```

```
##  [1] "batch6_Solid_4-1"  "batch6_Solid_5-1"  "batch6_Solid_7-1"
##  [4] "batch6_Solid_8-1"  "batch6_Solid_9-1"  "batch6_Solid_11-1"
##  [7] "batch6_Solid_13-1" "batch6_Solid_14-1" "batch6_Solid_15-1"
## [10] "batch6_Solid_19-1"
```

Clean list of 10 available samples:

```r
clean_names <- sub("-.*", "", sub("batch6_", "", glioma_with_solid[grepl("-1$", glioma_with_solid)]))
print(clean_names)
```

```
##  [1] "Solid_4"  "Solid_5"  "Solid_7"  "Solid_8"  "Solid_9"  "Solid_11"
##  [7] "Solid_13" "Solid_14" "Solid_15" "Solid_19"
```

```r
print(clean_names %in% targets$mapping) # these are all in the list of glioma SOLID patients with methy
```

```
##  [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

Method 2: assuming number following batch corresponds to SOLID patient ID (not sure if this is true)

```r
glioma_medip_rename <- sub("-.*", "", paste0("Solid_", sub("batch.*_", "", glioma_medip))) |> unique()
glioma_medip_rename
```

```
##  [1] "Solid_4"   "Solid_5"   "Solid_7"   "Solid_8"   "Solid_9"   "Solid_11"
##  [7] "Solid_13"  "Solid_14"  "Solid_15"  "Solid_19"  "Solid_119" "Solid_125"
## [13] "Solid_124" "Solid_122" "Solid_121" "Solid_120" "Solid_135" "Solid_134"
## [19] "Solid_133" "Solid_126" "Solid_140" "Solid_139" "Solid_137" "Solid_136"
## [25] "Solid_144" "Solid_143" "Solid_142" "Solid_141" "Solid_146" "Solid_145"
## [31] "Solid_17"  "Solid_182" "Solid_183" "Solid_80"  "Solid_18"  "Solid_147"
## [37] "Solid_148" "Solid_152" "Solid_157" "Solid_149" "Solid_207" "Solid_151"
## [43] "Solid_153" "Solid_184" "Solid_171" "Solid_154" "Solid_169" "Solid_158"
## [49] "Solid_156" "Solid_155" "Solid_20"  "Solid_161" "Solid_160" "Solid_159"
## [55] "Solid_166" "Solid_165" "Solid_164" "Solid_170" "Solid_162" "Solid_168"
## [61] "Solid_167" "Solid_163" "Solid_21"  "Solid_185" "Solid_186" "Solid_188"
## [67] "Solid_189" "Solid_190" "Solid_191" "Solid_22"  "Solid_23"  "Solid_24"
## [73] "Solid_174" "Solid_173" "Solid_175" "Solid_59"  "Solid_60"  "Solid_176"
## [79] "Solid_177" "Solid_172" "Solid_179" "Solid_178"
```

```r
table(glioma_medip_rename %in% targets$mapping)
```

```
##
## FALSE  TRUE
##    69    13
```

```r
# 13 of these renamed sampleIDs are in the list of glioma SOLID patients with methylation
```

```r
glioma_medip_rename[glioma_medip_rename %in% targets$mapping]
```

```
##  [1] "Solid_4"  "Solid_5"  "Solid_7"  "Solid_8"  "Solid_9"  "Solid_11"
##  [7] "Solid_13" "Solid_14" "Solid_15" "Solid_19" "Solid_22" "Solid_23"
## [13] "Solid_24"
```